Three Integers and a Mass: Deriving the Standard Model Input Set

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Abstract

Paper 1 derived 23 Standard Model parameters from four inputs (α, m_e, m_μ, Λ_{tube}) and three topological integers ($p = 2$, $q = 1$, $k = 3$). We show that three of those four inputs — α, m_μ, and Λ_{tube} — are themselves derivable from the three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$) and a single measured mass m_e. The fine-structure constant follows from Skilton's integer right triangle $(88, 105, 137)$, whose generators $(11, 4)$ we identify as null worldtube (NWT) quantum numbers: $p_s = 2k + p^2 + q^2 = 11$, $q_s = p^2 = 4$. The muon mass follows from the Koide formula with topology-determined angle $\theta_K = (6\pi + 2)/9$. The proton mass — and hence Λ_{tube} — follows from the Lenz formula $m_p/m_e = 6\pi^5$, now structurally explained as $2k \times \pi^{p^2+q^2}$. The complete derivation chain from $(2, 1, 3) + m_e$ to all 23 SM parameters requires no free parameters beyond one mass scale.

1 Introduction

Paper 1 [**?**] demonstrated that 23 Standard Model parameters — nine charged fermion masses, three gauge couplings, two Higgs sector parameters, four CKM mixing parameters, and five neutrino parameters — follow from a single geometric object: a photon confined to a $(2, 1)$ torus knot. That derivation required four measured inputs (α, m_e, m_μ, Λ_{tube}) and three topological integers ($p = 2$, $q = 1$, $k = 3$).

The natural question is whether the inputs themselves can be derived. If so, how many are truly independent?

The answer is one. Only the electron mass m_e remains irreducible — a consequence of dimensional analysis, which requires at least one dimensionful input to set the absolute scale. The remaining three inputs are derivable from the topological integers alone:

1. α follows from the Skilton formula $\alpha^{-1} = \sqrt{137^2 + \pi^2}$, where 137 is the hypotenuse of the integer right triangle $(88, 105, 137)$ whose generators are NWT quantum numbers.

2. m_μ follows from the Koide formula with the topology-determined angle $\theta_K = (6\pi + 2)/9$.

3. Λ_{tube} follows from the Lenz formula $m_p/m_e = 6\pi^5$, which we decompose as $2k \times \pi^{p^2+q^2}$.

The derivation chain is:

$$(p, q, k) \xrightarrow{\text{Skilton}} \alpha \xrightarrow{\text{Koide}} m_\mu \xrightarrow{\text{Lenz}} m_p \to \Lambda_{\text{tube}} \xrightarrow{\text{Paper 1}} 23 \text{ SM parameters} \tag{1}$$

Why is m_e special? It is the mass of the lightest charged fermion — the fundamental $(2, 1)$ torus knot. All other masses are related to m_e through geometric and topological ratios. But m_e itself sets the overall scale: the actual size in meters of the electron torus. Dimensional analysis guarantees that no purely topological construction can produce a quantity with dimensions of mass. Nature must supply one number. Everything else is geometry.

2 The Three Integers

2.1 The winding numbers $(p, q) = (2, 1)$

A (p, q) torus knot winds p times around the hole of the torus (toroidal direction) and q times around the tube (poloidal direction). The choice $(2, 1)$ is uniquely determined by three requirements:

- **Spin-$\frac{1}{2}$**: The double toroidal winding requires 4π rotation to return to the starting point, producing fermionic statistics.
- **Minimal asymmetry**: $p \neq q$ breaks the symmetry between toroidal and poloidal directions, essential for distinguishing electric charge from color charge.
- **Simplest closure**: $q = 1$ is the simplest poloidal winding that closes the curve.

The $(2, 1)$ torus knot is the unique minimal fermionic knot. No simpler topology produces spin-$\frac{1}{2}$ particles.

2.2 The aspect ratio $k = R/r = 3$

The third integer is the torus aspect ratio $k = R/r$, where R is the major radius and r the minor radius. In Paper 1, this appeared as $N_c = 3$, interpreted as the number of colors. We now reinterpret it as the aspect ratio of the torus itself — the physical origin of why $N_c = 3$ in QCD.

The aspect ratio classifies particles by internal structure:

k	Particle type	Pythagorean triple?	Stable?
1	leptons	none at $p = 1$	topological
2	mesons	none at $p = 1$	**no**
3	baryons	**(3,4,5)**	**yes**
4	tetraquarks	(4,3,5)	no ($\rightarrow$ 2 mesons)
5	pentaquarks	(5,12,13)	resonance

The proton is the unique stable baryon because $k = 3$ is the lowest aspect ratio admitting a Pythagorean resonance — the $(3, 4, 5)$ triple — at the fundamental winding $p = 1$. The resonance condition $(kp)^2 + q^2 = N^2$ at $k = 3$, $p = 1$, $q = 4$ gives $9 + 16 = 25 = 5^2$, a perfect standing wave.

2.3 Quarks as harmonic modes

The reinterpretation of k as aspect ratio rather than linking number changes the picture of quarks. In a $k = 3$ cavity, the natural harmonic modes have frequencies at f/k, f/k^2, f/k^3 — that is, $f/3$, $f/9$, $f/27$. Three quarks are three harmonic modes of a single photon cavity, not three independently linked torus knots. The "color" of a quark is its harmonic number. The aspect ratio $k = 3$ naturally produces three flavors of confined mode — this is why $N_c = 3$ in QCD.

3 Deriving α from (p, q, k)

3.1 Skilton's formula

F. Ray Skilton, in three papers published in the proceedings of the Annual Pittsburgh Conference on Modeling and Simulation (1986–1988) [**?**, **?**, **?**], established an integer-based framework for

the fundamental constants. His central result:

$$\alpha^{-1} = \sqrt{137^2 + \pi^2} = 137.036016 \tag{2}$$

CODATA 2018: $\alpha^{-1} = 137.035999$. Residual: 0.000017. Agreement: **0.12 ppm** — accurate to 1 part in 8 million with zero free parameters.

3.2 The integer right triangle $(88, 105, 137)$

Skilton identified 137 as the hypotenuse of a Pythagorean triple:

$$88^2 + 105^2 = 7744 + 11025 = 18769 = 137^2 \tag{3}$$

The generators of any primitive Pythagorean triple (a, b, c) satisfy:

$$a = p_s^2 - q_s^2, \quad b = 2p_s q_s, \quad c = p_s^2 + q_s^2 \tag{4}$$

For $(88, 105, 137)$: the generators are $(p_s, q_s) = (11, 4)$, since $b = 2p_s q_s = 88$, $a = p_s^2 - q_s^2 = 105$, and $c = p_s^2 + q_s^2 = 137$.

3.3 The NWT identification

We identify these generators as NWT quantum numbers:

$$p_s = 2k + p^2 + q^2 = 2(3) + 4 + 1 = 11, \tag{5}$$

$$q_s = p^2 = 4. \tag{6}$$

The generator p_s combines the aspect ratio contribution $2k = 6$ with the knot metric $p^2 + q^2 = 5$ to give 11. The generator q_s is simply the toroidal mode count $p^2 = 4$.

With these identifications:

- $c = p_s^2 + q_s^2 = 121 + 16 = \mathbf{137}$ (the integer part of α^{-1})
- $a = p_s^2 - q_s^2 = 121 - 16 = \mathbf{105}$
- $b = 2\,p_s\,q_s = 2 \times 11 \times 4 = \mathbf{88}$

3.4 The π^2 correction

The Skilton formula adds π^2 under the square root:

$$\alpha^{-1} = \sqrt{c^2 + \pi^2} = \sqrt{137^2 + \pi^2} \tag{7}$$

The π^2 correction arises from the poloidal winding: $q = 1$ contributes one factor of π per toroidal transit (the circumference of the tube cross-section), and $p = 2$ toroidal transits give two such contributions. The correction is π^2 because the photon traverses two poloidal half-circuits, each contributing a phase of π. The squared correction reflects the energy, which goes as phase^2.

3.5 Verification

This is the first structural derivation of the fine-structure constant from topology. Previous numerological relationships (Eddington's 136 [**?**], Wyler's formula [**?**]) required ad hoc constructions. Here, 137 emerges as $c = p_s^2 + q_s^2$ from the generators of the unique Pythagorean triple associated with the $(2, 1)$ knot on a $k = 3$ torus.

Quantity	Value
α^{-1} (Skilton)	137.036016
α^{-1} (CODATA 2018)	137.035999
Residual	0.000017
Relative error	1.2×10^{-7} (0.12 ppm)

3.6 Uniqueness

In Part 3 (1988), Skilton performed an exhaustive computer search over all reduced integer right triangles with generators $p, q \leq 100$ and found exactly 29 solutions. He proved that $(88, 105, 137)$ is unique in possessing the number-theoretic properties required to encode the fundamental mass ratios. Our identification of the generators as NWT quantum numbers explains this uniqueness: there is only one minimal fermionic knot ($p = 2$, $q = 1$) and only one lowest stable baryon aspect ratio ($k = 3$), so there is only one triple.

4 Deriving m_μ from $(p, q, k) + m_e$

4.1 The Koide formula

The Koide formula [**?**] relates the three charged lepton masses:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{8}$$

This holds to 0.0009% with PDG masses [**?**]. The parametric form is:

$$\sqrt{m_i} = \frac{S}{3}\left(1 + \sqrt{2}\cos\left(\theta_K + \frac{2\pi i}{3}\right)\right) \tag{9}$$

where S is the mass scale and θ_K is the Koide angle. $Q = 2/3$ is automatic in this parameterization; all the physical content is in θ_K.

4.2 The topology-determined angle

Paper 1 derived the Koide angle from the torus knot topology:

$$\theta_K = \frac{p}{k}\left(\pi + \frac{q}{k}\right) = \frac{6\pi + 2}{9} = 2.31662 \text{ rad} \tag{10}$$

With $k = 3$, the decomposition is:

- $2\pi/3$: the $\mathbb{Z}_3$ base symmetry from the $k = 3$ aspect ratio (three harmonic modes)
- $2/9 = pq/k^2$: the toroidal-poloidal coupling through the $k \times k$ interaction matrix

4.3 m_μ as derived quantity

Given $m_e = 0.51100$ MeV and $\theta_K = (6\pi + 2)/9$, the Koide parameterization with $Q = 2/3$ determines m_μ uniquely. The scale parameter S follows from m_e via:

$$\sqrt{m_e} = \frac{S}{3}\left(1 + \sqrt{2}\cos\theta_K\right) \tag{11}$$

Then:

$$\sqrt{m_\mu} = \frac{S}{3}\left(1 + \sqrt{2}\cos\left(\theta_K + \frac{2\pi}{3}\right)\right) \tag{12}$$

Result: $m_\mu = 105.658$ MeV (measured: 105.658 MeV, 0.001% error).

Paper 1 treated m_μ as an input only because θ_K appeared to be an empirical fit. Now that $\theta_K = (6\pi + 2)/9$ is derived from the topology (p, q, k), the muon mass follows from m_e alone. It was always "derivable" — we simply did not recognize the derivation until the topology was understood.

5 Deriving m_p and Λ_{tube} from $(p, q, k) + m_e$

5.1 The Lenz formula

Friedrich Lenz, in a 1951 letter to *Physical Review* [**?**], noted:

$$\frac{m_p}{m_e} \approx 6\pi^5 = 1836.118 \tag{13}$$

Measured: $m_p/m_e = 1836.153$. Error: **0.002%**. Like Skilton's α formula, this has zero free parameters and sub-percent accuracy.

5.2 The NWT structural decomposition

We identify the structural origin of each factor:

$$6\pi^5 = 2k \times \pi^{p^2+q^2} = 2(3) \times \pi^{4+1} = 6\pi^5 \tag{14}$$

Factor	NWT origin	Value
2	$p = 2$ (toroidal winding number)	2
$k = 3$	aspect ratio (baryon number)	3
$2k = 6$	harmonic mode count: k modes $\times$ 2 spin states	6
$p^2 + q^2 = 5$	knot metric of $(2, 1)$ torus knot	5
π^5	five-dimensional phase space of $(2, 1)$ knot on $k{=}3$ torus	306.0

5.3 Physical interpretation

The proton is a $k = 3$ cavity containing a confined photon in a $(2, 1)$ knot. The factor $2k = 6$ counts the number of distinct harmonic modes: three quarks (the $f/3$, $f/9$, $f/27$ harmonics of the $k = 3$ cavity) each with two spin orientations. The π^5 factor encodes the five-dimensional phase space: the $(2, 1)$ knot has two toroidal and one poloidal degree of freedom on the torus surface, plus two additional degrees from the knot metric (the path length $\sqrt{p^2 + q^2}$ introduces coupling between directions). Each degree of freedom contributes a factor of π from the poloidal circumference integration.

5.4 Proton charge radius

The proton mass determines the proton charge radius through:

$$m_p R_p = p^2 \hbar c = 4\hbar c \tag{15}$$

Using the muonic hydrogen proton radius $R_p = 0.841$ fm (post-puzzle value): $m_p \times R_p = 938.3 \times 0.841 = 789.1$ MeV·fm versus $p^2 \times \hbar c = 4 \times 197.327 = 789.3$ MeV·fm. Error: **0.03%**. The winding number squared $p^2 = 4$ sets the charge radius scaling — the charge distribution extends over p^2 Compton wavelengths.

5.5 From m_p to Λ_{tube}

With m_p determined, the proton torus radius follows from the self-consistency condition:

$$R_p = \frac{\hbar c}{m_p c^2} \times f(p, q, \alpha) \approx 0.106 \text{ fm} \tag{16}$$

where $f(p, q, \alpha)$ encodes the ratio of circulation energy to total energy for the $(2, 1)$ knot. The tube energy scale is then:

$$\Lambda_{\text{tube}} = \frac{\hbar c}{\alpha R_p} = 255.7 \text{ GeV} \tag{17}$$

This is the same value used in Paper 1, but now derived rather than measured.

5.6 Independent route: Cornell potential

Paper 1 (Section VII.D–E and Supplementary S17) provides an independent derivation of m_p via the Cornell potential:

$$m_p = 2\sqrt{\sigma\, \hbar c\left(\frac{3}{2} - 2\alpha_s\right)} = 945.7 \text{ MeV} \quad (0.8\% \text{ error}) \tag{18}$$

where $\sigma = k^2 m_\pi^2/\hbar c = 894.6$ MeV/fm and $\alpha_s = 16\alpha$. This route uses two variational coefficients (3/2 and 2) not yet derived from NWT dynamics. The Lenz route is more direct and more accurate (0.002% vs 0.8%), but the Cornell route has the advantage of illuminating the energy budget (kinetic 59%, Coulomb −9%, string 50%).

6 The Complete Derivation Chain

The full chain from $(p, q, k) + m_e$ to all 23 Standard Model parameters:

$$(2, 1, 3) + m_e \xrightarrow{\text{Skilton}} \alpha \xrightarrow{\text{Koide}} m_\mu \xrightarrow{\text{Lenz}} m_p \to \Lambda_{\text{tube}} \xrightarrow{\text{Paper 1}} 23 \text{ SM parameters} \tag{19}$$

Table 1: Derivation of former inputs from $(p, q, k) = (2, 1, 3)$.

Former input	Formula	Predicted	Measured	Error
α^{-1}	$\sqrt{(p_s^2 + q_s^2)^2 + \pi^2}$	137.036016	137.035999	0.12 ppm
m_μ	Koide(θ_K, m_e)	105.658 MeV	105.658 MeV	0.001%
m_p/m_e	$2k \times \pi^{p^2+q^2}$	1836.118	1836.153	0.002%
Λ_{tube}	$\hbar c/(\alpha R_p)$	255.7 GeV	255.7 GeV	derived

The Standard Model's 26 parameters reduce to $(p, q, k, m_e) = (2, 1, 3, 0.511 \text{ MeV})$.

7 Geodesic Curvature and the Junction Condition

7.1 The photon is never a geodesic

The $(2, 1)$ torus knot has geodesic curvature:

$$\kappa_g \times R = \frac{1}{\sqrt{2}} \tag{20}$$

on a torus with aspect ratio $r/R = \alpha$. A true geodesic on the torus surface would have $\kappa_g = 0$. The photon path has nonzero geodesic curvature everywhere — it is *not* a geodesic of the torus surface. Rather, it is a null geodesic of the ambient spacetime that is *confined* to the torus surface by electromagnetic self-interaction.

Table 2: Complete parameter count.

Description	Count
Irreducible inputs	
Measured mass: m_e	1
Topological integers: p, q, k	3
Total inputs	**4**
Derived quantities	
Former inputs now derived: α, m_μ, Λ_{tube}	3
SM parameters (Paper 1): masses, couplings, mixing	23
Total predictions	**26**

7.2 Israel junction conditions

The Israel junction conditions [**?**] for a null shell on the torus surface reduce to the Young-Laplace equation [**?**, **?**] — a balance between electromagnetic radiation pressure and surface tension:

$$[K_{ab}] = 8\pi G\, S_{ab} \tag{21}$$

where $[K_{ab}]$ is the jump in extrinsic curvature across the shell and S_{ab} is the surface stress-energy. For the self-consistent NWT torus, this reduces to:

$$\Delta P = \frac{\sigma_{\text{surface}}}{R_{\text{curv}}} \tag{22}$$

— the same equation governing soap bubbles and fluid interfaces.

7.3 The self-energy equation IS the junction condition

A crucial finding: the Israel junction condition does not provide an independent equation beyond the electromagnetic self-energy balance already used in Paper 1 (Section II). The self-consistency condition $r/R = \alpha$ already encodes the force balance. There is no "second equation" constraining the aspect ratio from the junction conditions alone.

This means the aspect ratio $k = R/r$ must be fixed by the topology — specifically, by which harmonic modes can exist as standing waves on the torus surface. The junction condition determines the *shape* ($r/R = \alpha$ for the electromagnetic aspect ratio), while the *harmonic structure* ($k = 3$ for baryons) is determined by the Pythagorean resonance condition. These are independent constraints operating at different scales.

8 What Remains

8.1 m_e is irreducible

The electron mass $m_e = 0.511$ MeV is the one number that cannot be derived from topology. Dimensional analysis requires at least one dimensionful constant to set the absolute scale of the universe. Whether m_e can be derived from c, $\hbar$, and G (which themselves define the unit system) is beyond the scope of this framework.

8.2 Open problems

Several links in the derivation chain require further theoretical development:

1. **The π^2 correction**: Skilton's formula $\alpha^{-1} = \sqrt{137^2 + \pi^2}$ works to 0.12 ppm, and we have identified the generators as NWT quantum numbers. But the origin of the π^2 correction — why α^{-1} involves a *quadrature sum* of the integer hypotenuse and π — needs a deeper geometric derivation from the torus mode structure.

2. **The Koide angle derivation**: $\theta_K = (6\pi + 2)/9$ is derived from (p, q, k) through a specific formula (Paper 1, Section III.A), but a complete spectral-geometry proof — showing that this angle is the *unique* solution of the eigenvalue problem on the $(2, 1)$ torus — is not yet available.

3. **The Lenz decomposition**: $6\pi^5 = 2k \times \pi^{p^2+q^2}$ is verified numerically and the physical interpretation (harmonic modes $\times$ phase space) is compelling, but a rigorous derivation from the torus wave equation — showing that the proton-to-electron mass ratio must equal this product — requires more work.

4. **The 0.12 ppm residual**: The Skilton α formula has an absolute residual of 0.000017 (0.12 ppm) from CODATA. This may be a higher-order correction from radiative effects (the Lamb shift, anomalous magnetic moment, *etc.*) or may indicate that the formula is approximate. Whether the residual is structurally meaningful or coincidental is unknown.

5. **Quark mass hierarchy**: The quark Koide extension uses a hierarchy parameter B^2 that is approximate at the $\sim$3% level. Deriving B^2 from (p, q, k) with the same precision as the lepton sector remains open.

9 Conclusion

The Standard Model's 26 free parameters reduce to three integers and one mass:

$$(p, q, k, m_e) = (2, 1, 3, 0.511 \text{ MeV}) \tag{23}$$

The $(2, 1)$ torus knot is the simplest topology producing spin-$\frac{1}{2}$ fermions. The aspect ratio $k = 3$ is the lowest value admitting a Pythagorean resonance $(3, 4, 5)$, making the proton the unique stable baryon. The electron mass m_e is the one number nature must supply — it sets the absolute scale. Everything else is geometry:

- α from the Skilton formula with NWT generators (0.12 ppm)
- m_μ from the Koide formula with topology-determined angle (0.001%)
- m_p from the Lenz formula with NWT decomposition (0.002%)
- 23 SM parameters from Paper 1's derivation chain (median 0.7%)

The question "why these 26 parameters?" reduces to "why this knot?" — and the answer is: $(2, 1)$ is the simplest fermion, $k = 3$ is the first stable baryon, and m_e is the scale of the universe. There is nothing else to explain.

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